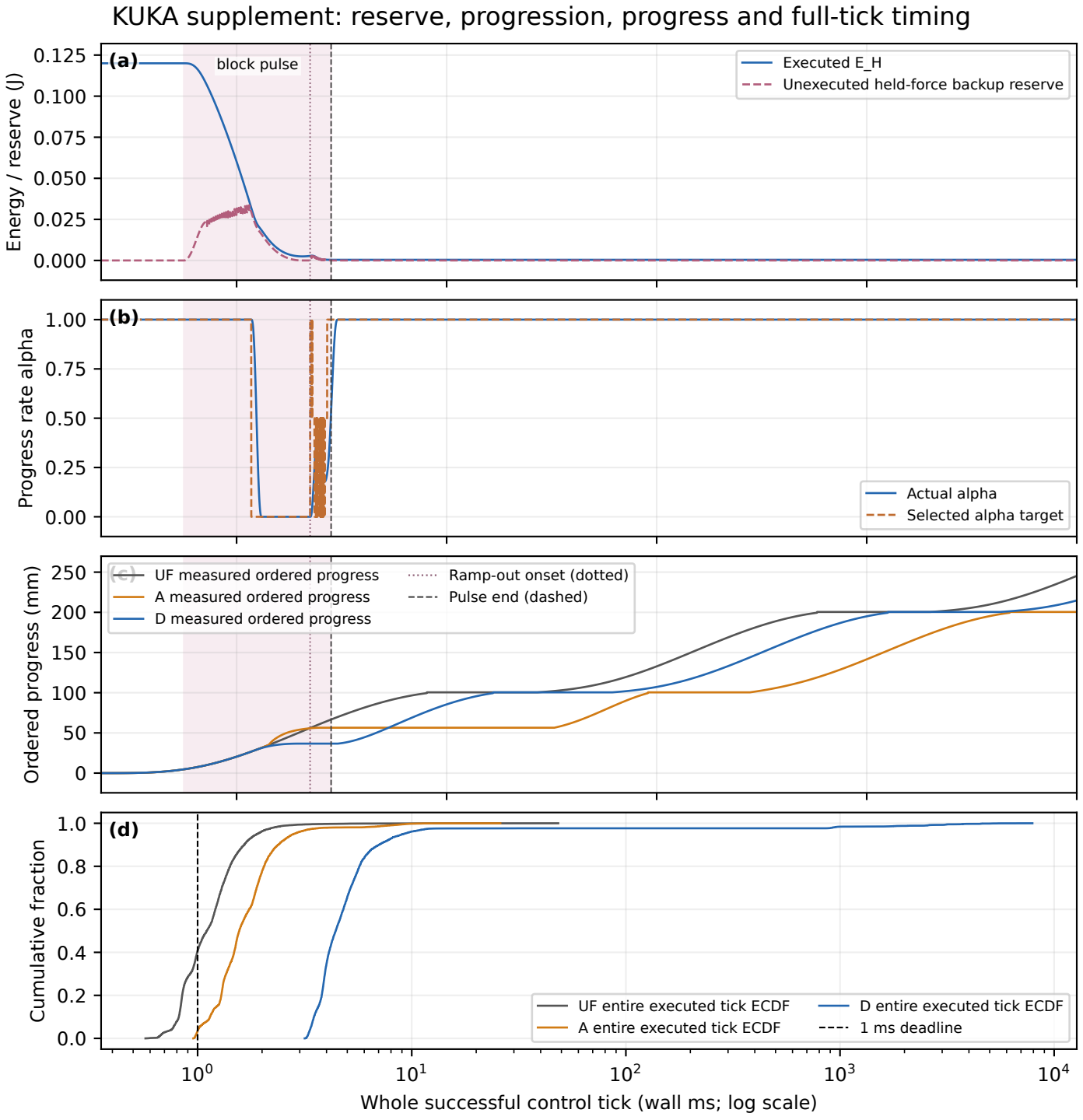

KUKA supplement: reserve, progression, progress and full-tick timing
(a)
block pulse
Executed E_H
Unexecuted held-force backup reserve
Energy / reserve (J)
0.125
0.100
0.075
0.050
0.025
0.000
(b)
Actual alpha
Selected alpha target
Progress rate alpha
1.00
0.75
0.50
0.25
0.00
(c)
UF measured ordered progress
A measured ordered progress
D measured ordered progress
Ramp-out onset (dotted)
Pulse end (dashed)
Ordered progress (mm)
250
200
150
100
50
0
(d)
UF entire executed tick ECDF
A entire executed tick ECDF
D entire executed tick ECDF
1 ms deadline
Cumulative fraction
1.0
0.8
0.6
0.4
0.2
0.0
10^0
10^1
10^2
10^3
10^4
Whole successful control tick (wall ms; log scale)